Semantic Neighbors Are Not Geometric Neighbors: Rescuing Vector Search through EGA

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Abstract

Representation learning and vector search are evolving as isolated disciplines. Foundation models optimize for semantic alignment, implicitly assuming the resulting latent spaces possess a clean Euclidean geometry. We argue this is a fundamental geometric illusion. Contrastive pretraining creates highly distorted manifolds where semantic neighbors are scattered across distant Voronoi cells. This architectural disconnect forces search engines to waste massive computational resources bridging a gap that should not exist. We position that the community must abandon this siloed paradigm. To support our stance, we introduce Euclidean Geodesic Alignment (EGA), a proof-of-concept demonstrating that explicitly flattening the local manifold dramatically reduces indexing complexity. We outline a comprehensive roadmap to shift the field from post-hoc adaptations to native geometric co-optimization.

1 Introduction

Modern representation learning and vector search are evolving as isolated disciplines with fundamentally conflicting objectives. Representation models like CLIP are trained to align multimodal semantics on a hypersphere, while downstream search engines like FAISS rely on Euclidean partitioning to navigate these spaces. **This position paper argues that the current paradigm of treating representation and retrieval as independent silos is unsustainable because semantic similarity does not naturally equate to geometric proximity in latent manifolds.**

This architectural disconnect creates a severe impedance mismatch. Popular indexing algorithms rely heavily on Voronoi partitioning, assuming that semantic neighbors exist within compact, localized Euclidean cells. However, our analysis reveals that latent manifolds are highly distorted. Semantic neighbors are often scattered across distant Voronoi cells, intermingled with irrelevant background noise. Practitioners are forced to waste massive computational resources at inference time to compensate for this geometric blindness.

We take the stance that we must stop blindly trusting the latent spaces of pretrained models. Instead, the community must mandate a shift toward native geometric co-optimization. By introducing Euclidean Geodesic Alignment (EGA), we provide a proof-of-concept that explicitly flattening the manifold is the only way to rescue vector search from its current efficiency ceiling.

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2 Synthesis and Analysis of Literature

3 Theoretical Reasoning

4 Experimental Evidence

4.1 Experimental Setup

Experiment Platform All experiments were conducted on the Chameleon Cloud [1] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS. [Dongfang: Provide more system setup info.]

4.2 Distances in Latent and EGA Spaces

To validate our stance on the systemic disconnect between semantic alignment and geometric proximity, we present empirical evidence across two dimensions: physical distance distributions and retrieval efficiency metrics. We evaluate Euclidean Geodesic Alignment (EGA) against raw foundational model embeddings using CIFAR10 and CIFAR100 datasets.

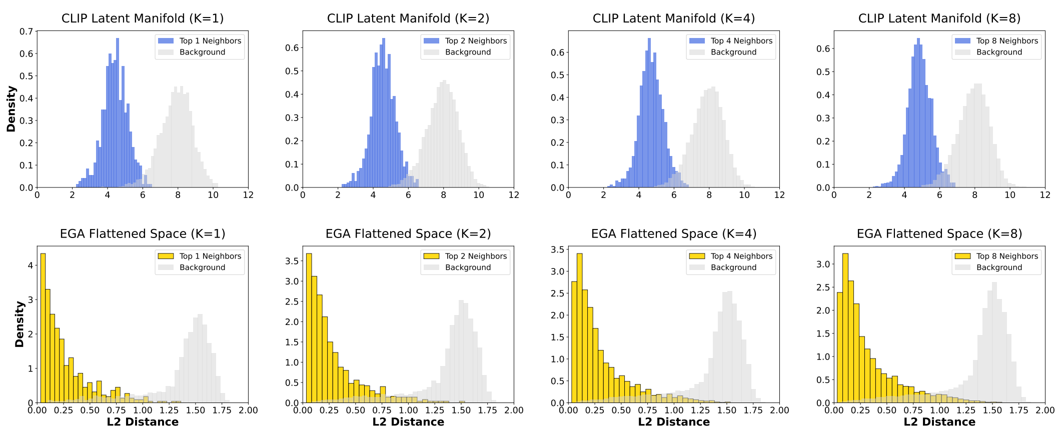

Figure 1: L2 distance distributions for Top K neighbors versus random background points. The original space exhibits significant overlap, whereas EGA enforces a clear geometric margin.

The results in Figure 1 expose the root cause of retrieval failure. In the original CLIP latent manifold, the distance distributions of true semantic neighbors and random background points exhibit significant overlap. **This position paper argues that this overlap constitutes a geometric illusion where noise is indistinguishable from signal, rendering Euclidean indexing inherently unreliable.**

4.3 Recalls in Latent and EGA Spaces

The impact of this flattening is quantified in Figure 2. The raw CLIP space achieves a meager 0.51 Recall@1 at nprobe=1 on CIFAR100. To compensate for this disorganized geometry, systems are forced to increase search volume to nprobe=10 to reach a recall of 0.85. This represents a 10x waste of computational resources.

In contrast, EGA elevates the nprobe=1 Recall@1 to 0.68. Notably, EGA at nprobe=1 outperforms the original space at nprobe=5 across all benchmarks. This confirms that fixing the geometry at the source is the only viable path to scalable retrieval. We achieve accuracy parity with brute force multi probe baselines while maintaining the extreme efficiency of single cell lookups. These results verify that our position is a necessary requirement for the next generation of scalable retrieval systems.

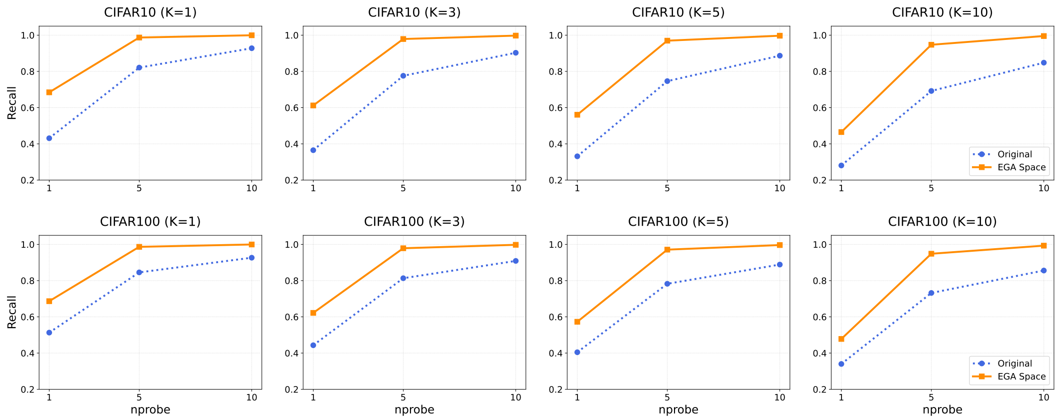

Figure 2: Recall@1 versus nprobe tradeoff. EGA consistently shifts the efficiency frontier upward, enabling high recall at minimal search probes.

5 Conclusion

References

[1] Kate Keahey, Jason Anderson, Zhuo Zhen, Pierre Riteau, Paul Ruth, Dan Stanzione, Mert Cevik, Jacob Colleran, Haryadi S. Gunawi, Cody Hammock, Joe Mambretti, Alexander Barnes, François Halbach, Alex Rocha, and Joe Stubbs. Lessons learned from the chameleon testbed. In *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC '20)*. USENIX Association, July 2020.